

Lecture – 7

Graphs –Part 2

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Spanning Tree

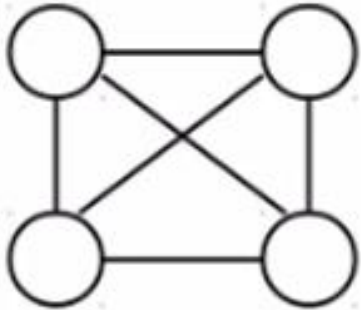
- A **Minimum Spanning Tree (MST)** is a fundamental concept in graph theory.
- A **spanning tree** of a **connected, undirected** graph is a subgraph that:
 - Connects all the vertices of the graph.
 - Has no cycles.
 - Contains exactly **$n - 1$ edges** (where n = number of vertices)
- **Minimum Spanning Tree (MST)**
 - Among all possible spanning trees of a weighted graph, the **MST** is the one with the **minimum possible total edge weight**.
 - If a graph has multiple spanning trees, there may be more than one MST if the edge weights allow.

Spanning Tree

- Suppose you have a connected, undirected graph.
 - **Connected**: every node is reachable from every other node.
 - **Undirected**: edges do not have an associated direction.
- A **spanning tree** of a graph is a connected subgraph in which there are no cycles.
- A spanning tree of a graph is cycle-free but still connects all the vertices.
- If G is a connected graph with n vertices and m edges, spanning tree of G must have $n-1$ edges, and no of edges deleted from G to get a spanning tree must be $m-(n-1) = m-n+1$

Question 1

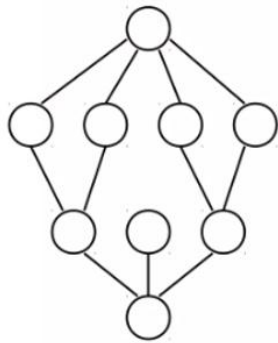
- Find all the spanning trees of the following graph.



A connected,
undirected graph

Question 2

- Consider the following undirected graph.



An undirected graph

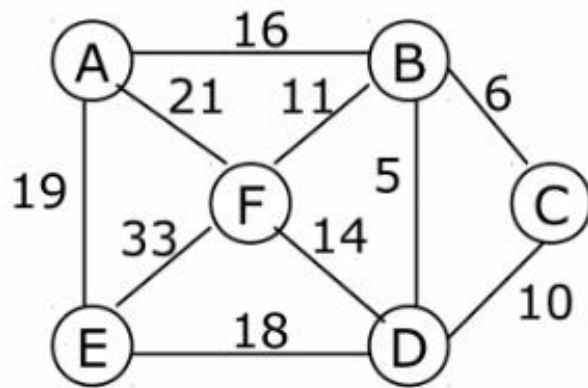
- Construct one spanning tree of the graph based on BFS and DFS traversal

Minimizing the Cost

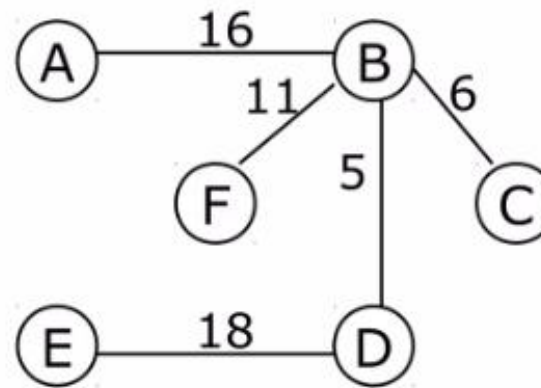
- Suppose you want to supply a set of houses (say, in a new subdivision) with:
 - electric power
 - water
 - sewage lines
 - telephone lines
- To keep costs down, you could connect these houses with a spanning tree (of, for example, power lines)
 - However, the houses are not all equal distances apart
- To reduce costs even further, you could connect the houses with a *minimum-cost* spanning tree

Minimum Cost spanning trees

- Suppose you have a connected undirected graph with a **weight** (or **cost**) associated with each edge
- The cost or weight of a spanning tree would be the sum of the costs of its edges.
- A **minimum-cost spanning tree** is a spanning tree of a connected undirected graph that has the lowest cost.
- If there are n vertices in the graph, then each spanning tree has n vertices & $n-1$ edges.



A connected, undirected graph



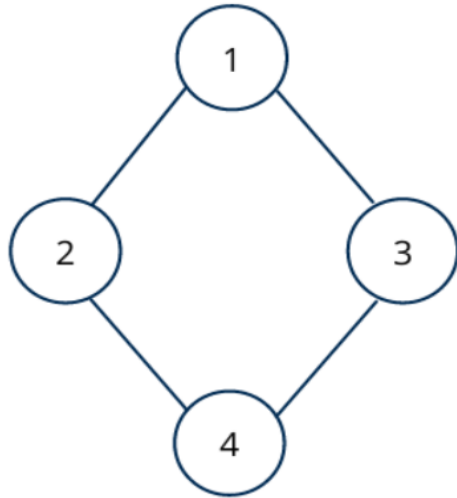
A minimum-cost spanning tree

Application of Minimum Spanning Tree

- Cluster Analysis
- Handwriting recognition
- Image segmentation
- Civil Network Planning
- Network Routing

Question 3

- Consider the following undirected connected graph:

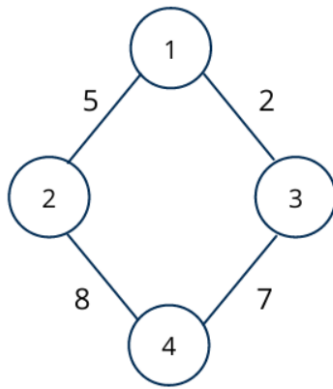


- Draw all possible spanning trees

Question 4

a) Define what is the minimum spanning tree?

- Consider the following graph:



b) List all possible spanning trees

c) Find the minimum spanning tree

Finding minimum spanning trees

There are two basic algorithms for finding minimum-cost spanning trees, and both are greedy algorithms i.e., these always process the edge with the least weight or cost.

- **Kruskal's algorithm:** Start with no nodes or edges in the spanning tree and repeatedly add the cheapest edge that does not create a cycle.
 - Here, we consider the spanning tree to consist of edges only.
- **Prim's algorithm:** Start with any one node in the spanning tree, and repeatedly add the cheapest edge, and the node it leads to, for which the node is not already in the spanning tree.
 - Here, we consider the spanning tree to consist of both nodes and edges

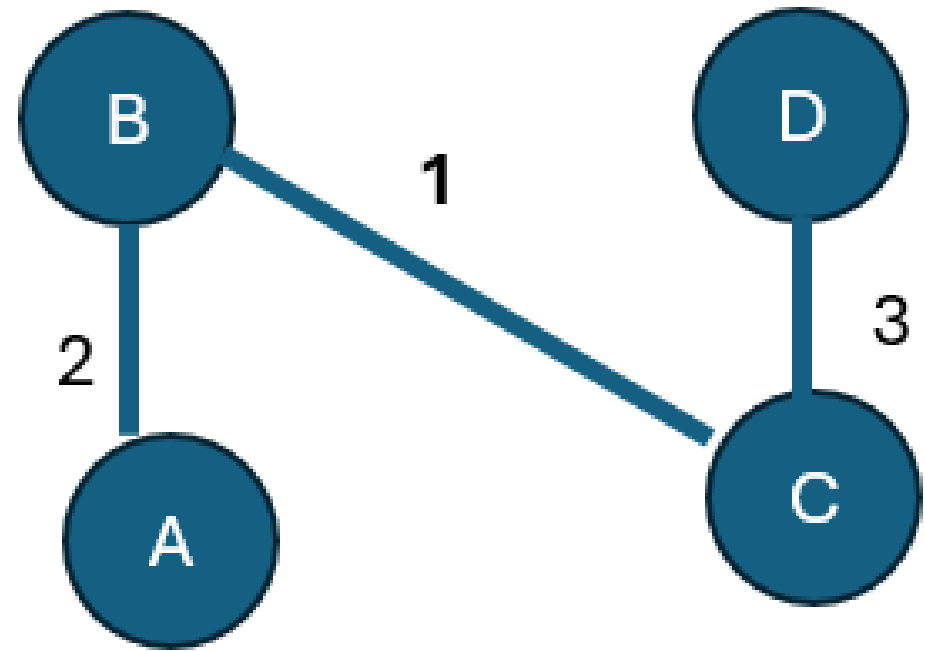
Kruskal's algorithm

- Joseph Kruskal was an American Mathematician, Statistician and Computer Scientist.
- He was born on January 29, 1928
- In Computer Science, his best-known work is Kruskal's Algorithm for computing the Minimal Spanning Tree (MST) of a weighted graph.

Kruskal's algorithm

- Kruskal's algorithm is an algorithm for computing MST of a weighted graph.
- A MST is spanning tree of connected, undirected graph which connect all the vertices together with the minimum total cost weightage of its edges and it always have a $V-1$ edges.

- Example



Pseudocode - Kruskal's algorithm

- The steps for finding MST using Kruskal's algorithm

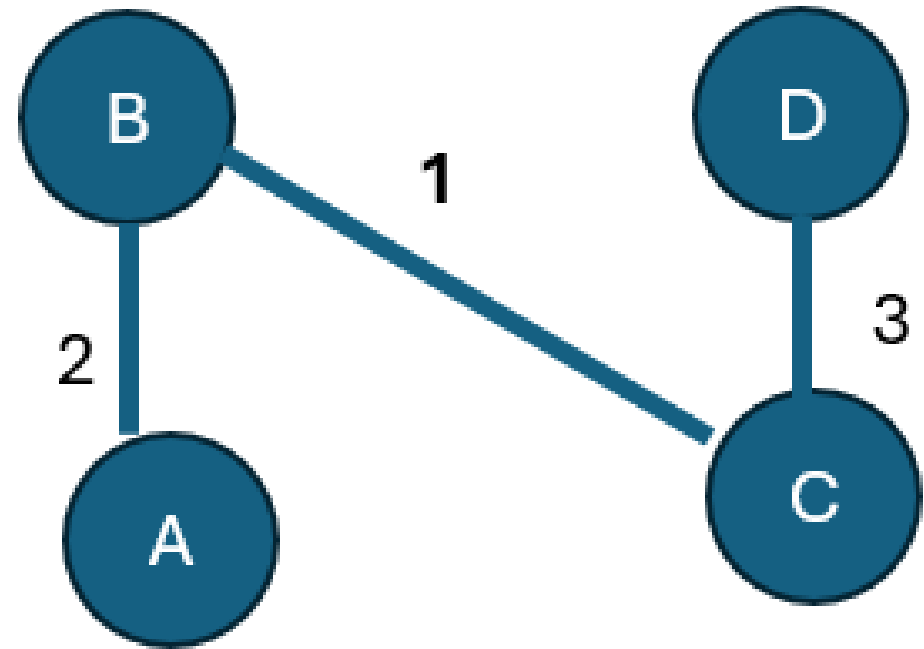
Step 1 : Sort all the edges in a non-decreasing order of their weight.

Step 2: Pick the smallest edge. Check if it forms a cycle with the spanning tree formed so far. If the cycle is not formed, include this edge. Else, discard it.

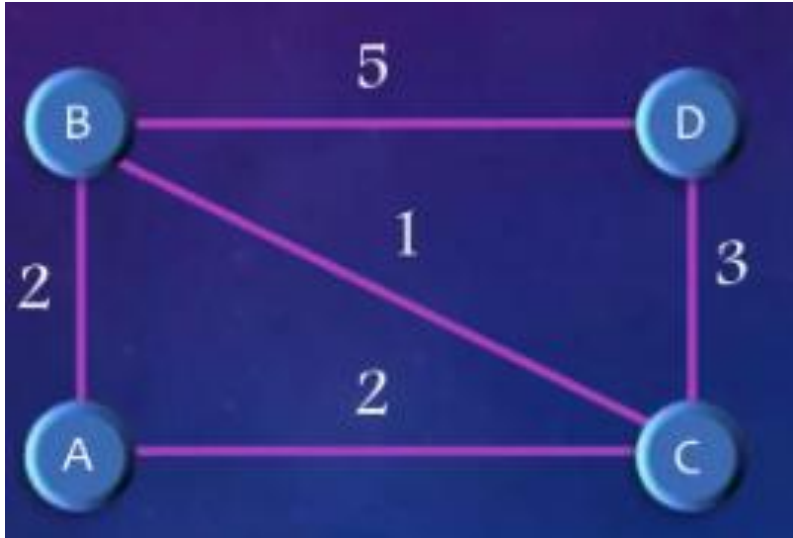
Step 3: Repeat step 2 until there are $(V-1)$ edges in the spanning tree.

How to find minimal spanning tree (MST) using Kruskal's algorithm?

- we will always start from the smallest weight edge and keep selecting edges that does not form any circuit with the previously selected edges.



Steps to find MST by using Kruskal's Algorithm



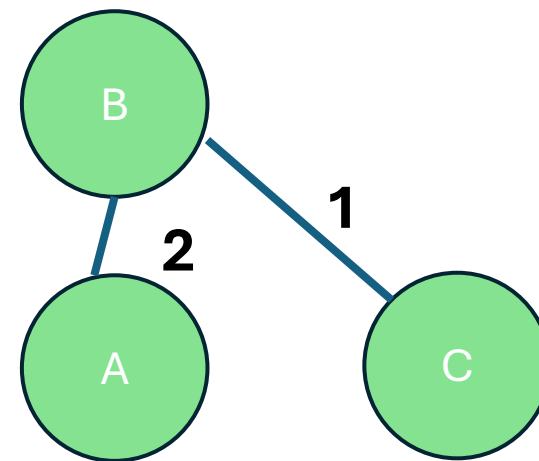
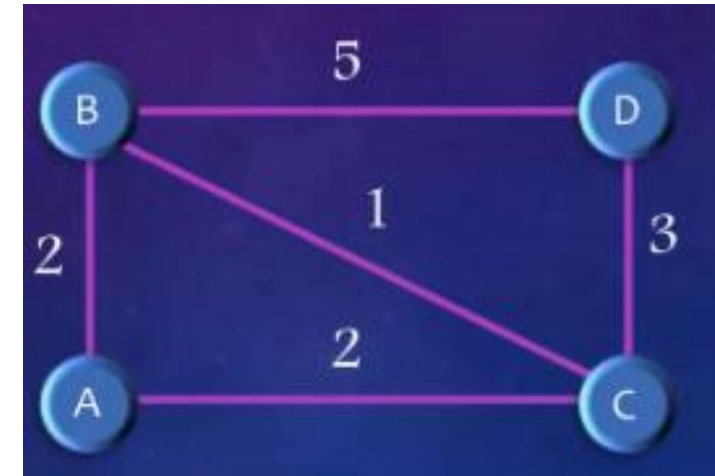
- We have five edges. So, our edge table will have 5 columns.
- Create the edge table (an edge table will have name of all the edges along with weight in ascending order)

- We will always start from the smallest weight edge and keep selecting edges that does not form any circuit with previously selected edges.
- Edge table

Edge					
Weight					

Steps to find MST by using Kruskal's Algorithm

- Fill the first column with the edge of minimum weight
- B-C is the minimum weight (weight=1)
- Next minimum weights are B-A (weight=2)
- **Pick next smallest edge: A-C (2)**
- **X skip**, it forms a cycle with A-B-C

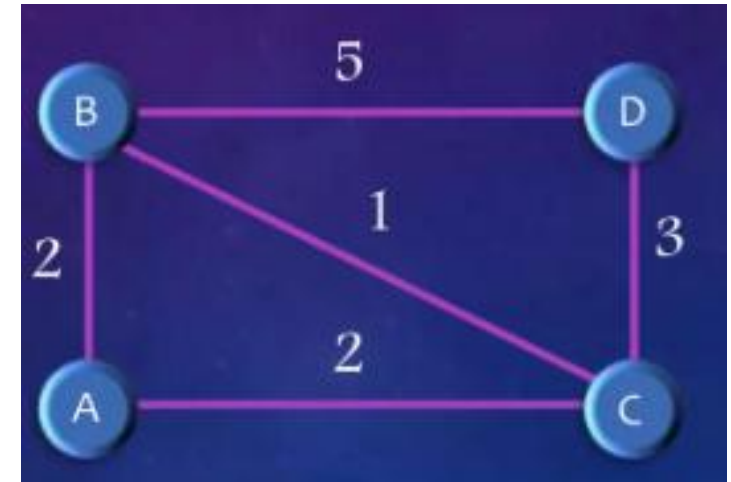


Edge	B-C	B-A				
Weight	1	2				

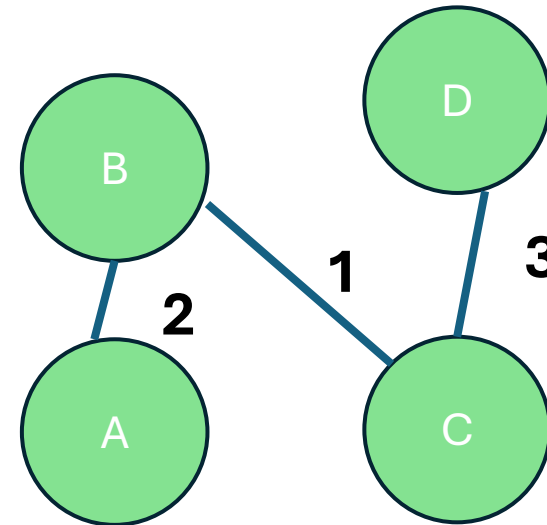
Steps to find MST by using Kruskal's Algorithm

- **Pick next smallest edge: C-D**
(Weight=3)

Edge	B-C	B-A	C-D			
Weight	1	2	3			

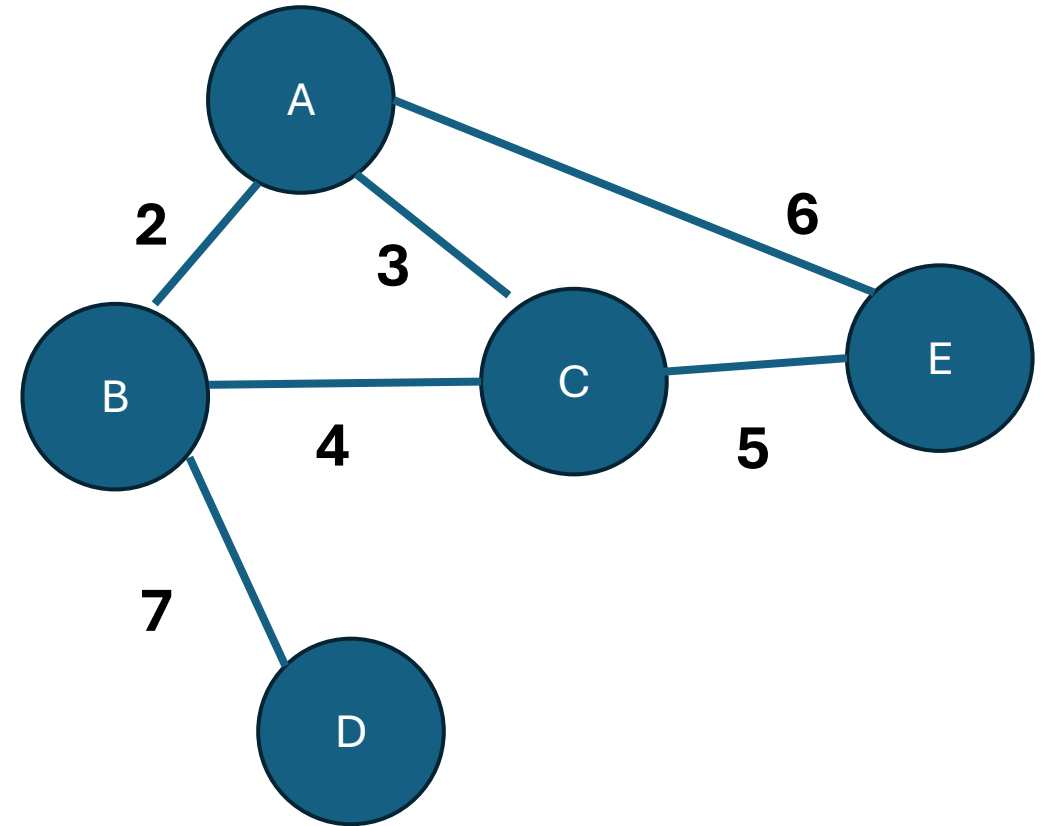


Total Minimum Weight =
 $1+2+3=6$



Question 5

- How to find minimal spanning tree (MST) using Kruskal's algorithm?



Prim's Algorithm for Minimum Spanning Tree (MST)

- Prim's Algorithm is a **greedy algorithm** used to find the **Minimum Spanning Tree (MST)** of a connected, undirected, weighted graph.
- A **spanning tree** is a subgraph that connects all vertices of the graph without forming cycles, and the **minimum spanning tree** is the spanning tree with the minimum possible total edge weight.
- Deference between the **Prim's Algorithm** and Kruskal's Algorithm
- **Kruskal's Algorithm:** *Sort all edges by weight and keep adding the cheapest edge that doesn't form a cycle*
- **Prim's Algorithm:** *Start from one node and grow the MST step by step by adding the cheapest edge leading out.*

Steps of Prim's Algorithm

Step 1 : Initialize

- Declare an array visited[] (or a set) to keep track of visited vertices.
- Select an arbitrary root vertex, say S, and mark it as visited.
- Initialize the MST result as empty.

Step 2 : Find Minimum Edge

- From all edges that connect **a visited vertex** to **an unvisited vertex**, choose the edge with the **minimum cost**.

Step 3 : Update MST

- Add this least-cost edge to the MST.
- Mark the newly connected vertex as visited.

Step 4 : Repeat

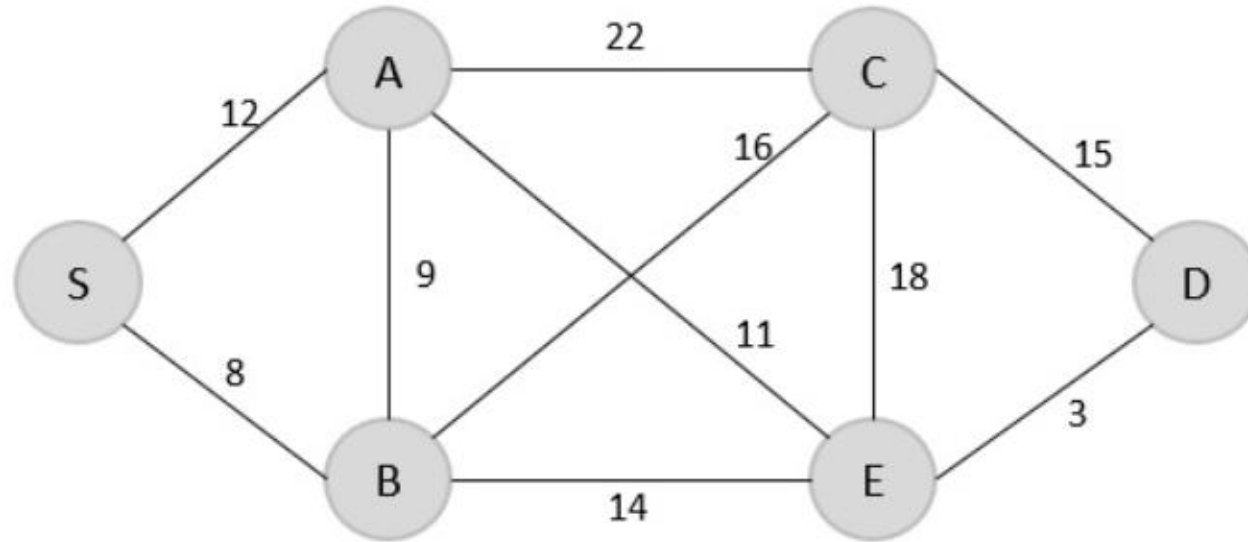
- Repeat steps 2 and 3 until all vertices are visited (i.e., MST contains $V - 1$ edges for a graph with V vertices).

Step 5 : Calculate Cost

- Sum the weights of the edges in the MST to get the total minimum cost

Question 6

- Consider the following graph



- Find the minimum spanning tree using prim's method (greedy approach) for the graph given below with S as the arbitrary root.
- Write a C program to find the Minimum Spanning Tree (MST) using Prim's Algorithm